

“It Must Be a System Thing:” Information Infrastructure Genres as Sites of Inequity

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ABSTRACT

Drawing on qualitative data collected from program participants in the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), I show how federal government assistance information infrastructure often does not remediate, and instead exacerbates, existent inequalities. I use the example of WIC’s Approved Product List (APL) to show how the APL, as a genre that’s part of WIC’s information infrastructure, contributes to a hyper-standardized benefit redemption process that increases visibility and vulnerability for program participants. This article argues that increased attention to the genres that make up information infrastructures may help to better locate sites of inequity like the APL, and better understand how systemic/structural problems perpetuate infrastructurally.

CCS Concepts

Information Systems

Keywords

Genre, Infrastructure, Inequity, Access

Victoria, an African-American single mom of two, lives in Amherst, New York. She’s been enrolled in WIC twice: first in 2014, then in 2019. Before the pandemic, and now, she has had a difficult time buying food with her Special Supplemental Nutrition for Women, Infants, and Children (WIC) benefits at the grocery store. She has frequent issues that are altogether mystifying to her. The visibility of her issues at the cash register reveals her status as a WIC participant, creating a negative and stressful grocery store experience:

When you’re Black you have this feeling like, the people in the line and the cashier, they’re all judging you and they’re thinking like, she’s on this program. This must be food stamps or something. I’ve had people say like, “What are you doing? What are those? How are you paying for this food?” Like pointing it out to me saying, like, making it seem like I’m taking advantage of a system. It’s a little dehumanizing. It’s such a long process.

And sometimes I feel like the sideshow at the store. You are told, hey, that’s not the right brand. It’s not the right size. This is everything that fits in this category but for some reason it’s not being counted. This was .1 of an oz over, so now you’re being charged for it. There’s so many times where I have problems.

No grocery trip is the same. It feels very much like a “powers-that-be” situation that dictates it. When you try to figure it out for yourself, like thinking ‘I need these items, in these sizes,’ then you go to the cashier and they don’t work in the system, and you ask the cashier, and the cashier doesn’t know. Then the cashier asks the manager, and the manager doesn’t know. Well, let me try to call WIC. And then WIC doesn’t know. So it’s like, it must be a system thing.

This article posits that information infrastructure genres are sites to investigate systemic problems like the “system things” that Victoria describes and other WIC participants experience. Drawing from an infrastructural inversion (Bowker & Star, 1999) of WIC’s information infrastructure, as well as case examples of WIC

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participants navigating this infrastructure, this article provides preliminary findings concerning the role of one genre within WIC's infrastructure, the Approved Product List (APL), in shaping access for WIC participants.

This study found that although APLs were designed as part of a wider effort to “streamline” and “modernize” WIC through implementing Electronic Benefit Transfer (EBT) (“Making WIC Work Better”), APLs create a hyper-standardized, high-stakes environment in which program participants must decide whether to advocate for themselves to successfully redeem their benefits and increase their visibility as someone “on welfare” or leave the store empty-handed. Ultimately, these preliminary findings suggest that the genres that make up information infrastructures may be key sites to better understand how systemic racism, ableism, and classism and their intersections perpetuate.

LITERATURE REVIEW

This study of the APL extends from the recent infrastructural “turn” in writing studies and technical communication (DeVoss et al., 2005; Grabill, 2010; Frith, 2020; Hart-Davidson et al., 2007; Read, 2019; Swarts, 2010), taking up Read's (2019) call to “make visible the boring, yet essential, things that constitute so much of organization life yet are largely invisible to it” (p. 262) through better understanding the infrastructural function of the APL. Understanding what Read termed the “infrastructural function” of writing includes taking into account the following:

1. Inclusiveness: a broad scope for what counts as writing;
2. Relationally defined: a focus on what writing does for something or someone (incorporates rhetorical genre theory);
3. Alliance brokering: writing that mediates essential alliances; and
4. Mission critical: writing that is essential to the operations of an organization. (p. 246)

As Read (2019) explained, “the infrastructural function of writing is defined by what writing does for something or someone rather than by the formal characteristics ... of any given document or writing activity ... the infrastructural function, then, must be able to identify for what or whom the writing functions as infrastructural” (p. 251). Drawing on rhetorical genre theory, and utilizing Read's elements of the infrastructural function of writing, made it possible to analyze the APL as a genre that is part of WIC's information infrastructure, then use that analysis as contextualization for the case examples of WIC participants' experiences in the program. Infrastructure is at once universal and highly situated. Though it is certainly useful to understand the complex web of WIC's information infrastructure, this analysis alone cannot capture how individual WIC participants experience this infrastructure as they navigate grocery shopping and checkout as they attempt to redeem their government benefits. On a given day, at a specific grocery store, with particular cashiers and managers on duty, WIC participants experience information infrastructure's mediating influence differently, for better and worse.

WIC is a national program that “provides Federal grants to States for supplemental foods, health care referrals, and nutrition education for low-income pregnant, breastfeeding, and non-breastfeeding postpartum women, and to infants and children up to age five who are found to be at nutritional risk” (USDA). WIC serves nearly

half of all infants born in the United States. Increased access to food, information, and healthcare is central to WIC's mission, with the specific goal to resolve participants' food insecurity and nutritional risk, promoting an overall security (both in their ability to attain food and maintain a healthy lifestyle). In order to enable people to enroll in WIC, receive their assigned benefits, and redeem those benefits at the grocery store, there is a vast network of people, processes, procedures, tools, facilities, and technology which “[support] the creation, use, transport, storage, and destruction of information” (Pironti, 2006)—WIC's information infrastructure. Woven throughout WIC's information infrastructure are genres that outline program guidelines, processes and procedures, and document information that circulates within the program (Figure 1).

Genres are, as Miller (1984) described, “typified rhetorical actions based in recurrent situations” (p. 159)—the genres that make up WIC's information infrastructure, likewise, are routinized and help create, use, transport, store, and destroy information across WIC's federal office and state agencies.

The Approved Product List (APL) is a critical genre that functions within WIC's information infrastructure, documenting every eligible item a WIC program participant can purchase using their benefits. At face value, the APL resists direct analysis (see Figure 2). It is mundane and highly technical—a massive Excel spreadsheet filled with brand names, item sizes, and categorical markers. Each WIC state agency maintains their own APL that documents the Universal Product Code (UPC), Item Description, Category Code, Category Description, and Subcategory Code for all eligible food items available for purchase through approved vendors (typically grocery stores) in their state. APLs enable WIC state agencies to document eligible food items in a central location through hyper-standardization. In previous iterations of WIC's information infrastructure, eligible food items were determined by price caps, but APL documentation generates individual approvals for specific food items. Each WIC state agency has an APL manager who is responsible for maintaining the APL. Additionally, each state agency has a small committee of staff workers who are responsible, along with the Director, in setting guidelines for how “strict” of an APL the state should have (Jovi, personal interview). Some states opt for “stricter” and “leaner” APLs that are a smaller file size, easier to manage, and easier to update, whereas other states opt for sprawling, messier APLs that try to more accurately capture what food items are WIC-eligible. Though there are centralized processes at the national level for approving WIC foods, there is significant variation in how states manage this approval, and this variation is reflected across the APLs.

Therefore, the APL is a recurring genre, even given the variation across states. APLs are stored in an online repository called the National Universal Product Code (NUPC) database, managed by the United States Department of Agriculture (USDA). A given state's APL may be less up-to-date or extensive than another's, depending on how often it is updated and re-uploaded to the NUPC database. Although the NUPC can only be accessed by WIC state and federal administrators and staff, copies of APLs are publicly available on state WIC websites for vendors to reference.

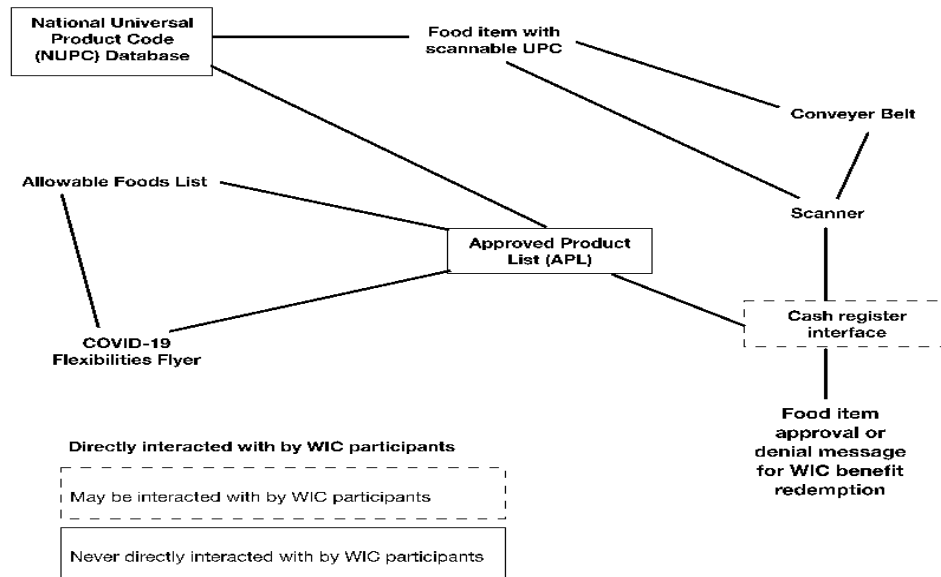


Figure 1. Genre ecology map of WIC benefit redemption.

Missouri WIC Approved Product List

UPC	Item Description	Cat Code	Category Description	Subcat Code
000651003214	Ocean Mist Romaine Hearts 22 oz	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	001
000651011004	Head Lettuce	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	001
000651111025	Ocean Mist Farms Broccoli 1 Bunch	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	001
000651111117	Cauliflower	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	001
000651511016	Ocean Farms Celery 1 Stalk	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	001
000651511030	Ocean Mist Farms Celery Hearts 1 Lb	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	001
000651961026	Ocean Mist Organic Celery	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	001
000651961040	Ocean Mist Organic Iceberg Lettuce	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	001
011110001320	Kroger 100% Whole Wheat Roundtop Bread 16 oz	16	BREAD/WHOLE GRAINS	001
011110015747	Kroger Whole Wheat Tortilla 16 oz	16	BREAD/WHOLE GRAINS	008
011110016188	Kroger Creamy Peanut Butter 16 oz	06	LEGUMES	001
011110016195	Kroger Crunchy Peanut Butter 16 oz	06	LEGUMES	001
011110017918	Simple Truth FRZ Organic Cherry Berry Blend 2 Lbs	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	002
011110017987	Kroger Shredded Whole Milk Mozzarella 8 oz	02	CHEESE/TOFU	001
011110018342	Kroger Grape Juice 64 oz	54	JUICE - 64 OZ	002
011110018359	Kroger White Grape Juice 64 oz	54	JUICE - 64 OZ	002
011110024985	Simple Truth Organic Slice Baby Bella Mushrooms8oz	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	001
011110024992	Simple Truth Organic Baby Bella Mushrooms 16 oz	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	001
011110026309	Dillons 100% Whole Wheat Roundtop Bread 16 oz	16	BREAD/WHOLE GRAINS	001
011110029065	Kroger Original Oats Instant Oatmeal 12 oz	05	BREAKFAST CEREAL	001
011110032522	Private Selection FRZ Sliced Strawberries 3 Lbs	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	002
011110035523	Simple Truth Organic FRZ Leaf Spinach 12 oz	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	002
011110035530	Simple Truth Organic FRZ Edamame 12 oz	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	002
011110035547	Simple Truth FRZ Organic Broccoli Cuts 12 oz	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	002
011110035554	Simple Truth FRZ Organic Mukimame 12 oz	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	002
011110035561	Simple Truth FRZ Organic Corn 12 oz	19	FRUIT & VEGETABLE CASH VALUE BENEFIT	002

Figure 2: Excerpt of Missouri's Approved Product List (APL)

As a genre, an APL enables WIC program participants social action by permitting benefit redemption within the Electronic Benefit Transfer (EBT) system. EBT is new to WIC, replacing a paper check system for benefit redemption. In an EBT system, WIC program participants redeem their benefits using an eWIC card, rather than paper checks, and all eligible WIC items are documented on the APL. Before EBT, eligible WIC items were processed through standardized price caps. The “paper check system” was criticized as a cumbersome, stigmatizing process for WIC participants to experience, and the introduction of EBT was intended to streamline and modernize benefit redemption (“Making WIC Work Better”).

Though it was apparent from an initial exploration of WIC’s information infrastructure that the APL documented eligible WIC items, it was unclear if (or how) the APL influenced the experiences of WIC participants as they grocery shopped and attempted to pay for groceries using their WIC benefits. I set out to better understand how the APL, as a genre within WIC’s information infrastructure, “worked.”

RESEARCH DESIGN

In this article, I work to answer the following research questions:

How do genres that make up information infrastructures shape access to digitally-mediated government benefits?

How can an infrastructural perspective reveal inequities that result from a genre that is part of an information system that mediates a process most often engaged with by vulnerable populations?

To answer these questions, I engaged Bowker and Star’s (1999) methodology of infrastructural inversion to contextualize interviews conducted with WIC participants within an infrastructural perspective. Infrastructural inversion is a method developed by information scientists Bowker and Star to bring to the forefront of analysis the classification and standardizations that make up information infrastructure. Bowker and Star have suggested tracing the “cumulative mess” of a given system, a “reverse engineering [that] reveals the multitude of local political and social struggles and compromises that go into the constitution of a ‘universal’ classification” (p. 48). Information infrastructure development and management takes place within institutions and is, therefore, not readily identifiable or visible to those utilizing technology. However, the information infrastructure defines and influences the genres that extend from that system. The standards, protocols, and other technical documents that mediate and maintain the information infrastructure are markedly invisible to everyday people who are affected by them. Infrastructural inversion uncovers how people are served (and not served) by infrastructure. For example, Read’s (2019) study of a supercomputing center revealed that the interpretation of the inscriptions produced by the supercomputing machine shaped the work of the operations team and served an infrastructural function for the operational annual review (OAR) report (“The Infrastructural Function”).

Following this infrastructural inversion, I conducted interviews with WIC program participants (IRB # STUDY00145763, University of Kansas, 2019). The interview data, contextualized within the data collected from infrastructural inversion, helped me understand how WIC participants navigate their grocery shopping experiences mediated by the APL as it operates behind-the-scenes. This study is ongoing, but the case example participants shared in

this article are included below (Table 1).

I utilized convenience sampling as the method for recruiting participants for this study by reaching out to self-identified WIC participants on Facebook. These WIC participants posted publicly about their experience on WIC, either on their local WIC page, the national page, or through a status update. Convenience sampling has several affordances for my project, particularly under the constraints of conducting qualitative research during the pandemic (2020-2021). I could reach out to a large number of WIC participants, staff, and vendors across the United States without any kind of initial “filter” that ruled out particular participants, staff and administration, or vendor representatives. Convenience sampling allowed me to make the best of the situation, get in contact with who I could, and work to answer my research question through a flexible means of data collection.

Participant Pseudonym	Race/Ethnicity	Program Status	Nutritional Risk Status	Family Size	State of Residence	Time in Program (in years)
Lilah	White, NH	PP**	Diet-based	4	NJ	1.5
Lavender	White, NH	P****	Medical (Disabled)	2	NH	0.5
Victoria	Black, NH	PP	Diet-based	4	NY	4
Harmony	Black, NH	PP	Diet-based	4	KS	3

*Non-Hispanic
 **Post-partum
 ***Breastfeeding (partially or full)
 ****Pregnant

Table 1: WIC Participant Case Examples Demographic and Geographic Information

FINDINGS

This study found that the APL is the central genre that enables (or prohibits) a WIC program participants’ benefit redemption, even though WIC participants never interact directly with it. If a WIC participant attempts to purchase an item using their WIC benefits, and it is not listed on the APL, it will not be approved. Thus, state agencies that maintain and update their APL must attempt to keep this documentation as accurate as possible. Each state has its own APL and is responsible for updating it to reflect food item eligibility. States must follow national guidelines for food item eligibility (including nutritional value, cost, and availability), and must review every food item against these guidelines before adding it to their APL. If an eligible food item documented on the APL changes in size (even by an ounce), it must be removed from the APL, re-submitted for review, and approved before it appears on the APL again. One WIC director, Jovi, described in her interview that “it’s a constant game of cat-and-mouse trying to keep the APL updated in our system.” Put simply, if a food item is documented on the APL, when scanned by a cashier, it will be validated as a WIC-eligible purchase, but if it is absent from the APL, it will be rejected, even if it is indeed a food item that is eligible to purchase through WIC.

This hyper-standardization creates a high-stakes environment WIC participants must navigate during grocery shopping checkout, even though they have no knowledge of the APL’s existence. On hand, WIC participants may have other documentation genres that outline eligible WIC purchases (Figure 3). They may have their Allowable Foods List (in digital or printed form), previous receipts with their available listed balance, and, during the pandemic, may also bring along a flexibilities flyer that outlines eligible substitutions for certain food items. They place their items on the conveyer belt, sorting out WIC items from items they intend to purchase out of

pocket. They swipe their eWIC card and enter their PIN while the cashier scans the items' Universal Product Code (UPC). When an item's UPC is scanned, the item is first compared to the available benefit balance. For example, if a WIC participant has already purchased four gallons of 1% milk and tries to purchase another, the EBT system will flag this item as ineligible.

Second, the item is compared to the APL. If a participant tries to purchase a gallon of 1% milk that isn't documented on the APL, the EBT system will flag this item as ineligible as well.

Critically, a store's current APL isn't necessarily the most up-to-date or correct APL. Stores must manually "pull down" (download) the nightly file of the APL sent from the WIC state agency to their cash registers every day. Even if stores pull down the nightly file of the APL, it still may not be up to date, depending on how recently a state agency has reviewed eligible items for each vendor, as well as processed requests for items to be added.

That the APL functions this way is universal across WIC. However, WIC program participants experience this infrastructure differently, across different grocery stores, APL versions, and unique interactions with grocery store staff. Victoria, Lavender, Lilah, and Harmony's experiences, as case examples, shows how the APL as a genre operationalizes uniquely for individuals, shifting even day-to-day or store-to-store.

Victoria

Victoria, whose experiences grocery shopping and using her WIC benefits were highlighted at the opening of this article, feels hyper-visible during checkout, even when things go right. During our interview, Victoria described that although using an eWIC card was generally "discreet" and can lead to a "normal transaction" where "[she] doesn't have to feel like a sideshow that is holding up the line," most often, problems occur. For example, at the beginning of the COVID-19 pandemic, WIC vendors with limited WIC-eligible items in stock offered "flexibilities" that allowed participants to substitute items with other brands if the original item was out of stock. When Victoria tried to make a substitution, however, she described that she was unsuccessful:

Actually going into the store and attempting to get different sizes and more choices, using the updated Allowable Foods List, and seeing that there are more choices for cereal ... they said they don't have them. So I can get more choices, but they aren't available. It says on the List we have these expanded options, but when you actually try to check out it doesn't register in their system. So it's like, okay, well, there's nothing cashiers can do about that. And there's nothing I can do about that, and even if I call WIC there's nothing they can do about that. So okay, I guess I'll just pay for this box of cereal, which is pretty inconvenient especially since the List says I can get it.

Although Victoria only mentions the Allowable Foods List as a consequential genre in her attempt to redeem an eligible substitution, the APL is also at play here. The APL is the primary mechanism for individual food item approval or denial during checkout via a matching UPC, or a missing UPC. Because of inconsistent and limited items in stock, particularly during periods of "runs" on grocery stores by panicked shoppers throughout 2020, the APL created further constraints for WIC participants as they attempted to find and purchase eligible items. As the pandemic

wore (wears) on, the USDA created a series of WIC COVID-19 "flexibilities" that have been approved for the duration of the public health emergency (USDA/Food and Nutrition Services, n.d.). These flexibilities apply to the full spectrum of WIC activities, from implementing digital enrollment, telehealth visits, remote issuance of benefits, food package flexibility, and options for food package distribution. These flexibilities are essentially a workaround of WIC's current information infrastructure that was designed for "in-person" interactions. The increased flexibility of food packages is intended to "permit appropriate substitutes" for the types and amounts of certain WIC-prescribed foods if their availability is limited. For example, some states have been approved to allow participants to:

- Substitute milk of any available fat content if prescribed varieties are not available;
- Substitute authorized whole grains in package sizes up to 24 oz. when 16 oz. packages are not available; and
- Substitute 18-count cartons of eggs when 12-count cartons are unavailable ("USDA/Food and Nutrition Service" 2021).

These waivers are intended to expand what is eligible within the WIC program without having to make direct changes to the APL; importantly, these items are only eligible when eligible items are unavailable. The flyers are meant to override the dictates of the APL: for example, if 12-count cartons of eggs are unavailable, 18-count cartons become eligible, even though when scanned they will be rejected as ineligible. This flexibility, at the same time that it promotes an increased range of food item eligibility, also places additional strain on the interaction between cashiers and WIC participants during checkout. As Jovi, a WIC Director in Montana, explains, this flexibility "positions the cashier as the police" who determine whether or not items are out of stock, and therefore if the alternative is eligible (personal interview). Cashiers must decide whether or not they're going to accept the waiver as authoritative over the authority of the APL. These waivers work to act as loopholes for the APL, and in doing so, reveal that the APL was designed under the assumption that WIC-eligible items would always be in stock and that substitutions would ultimately not be necessary for WIC participants to fully redeem their benefits. This assumption becomes visible under the changing conditions introduced by COVID because the APL no longer works as it was originally designed. For Victoria, this added a frustrating pattern to an already difficult process—not only were substitutions also unavailable at times, but sometimes substitutions weren't honored by cashiers (often not up-to-date with WIC's latest policies) during checkout.

As a Black woman, Victoria is, as Jovi describes, "policed" by the cashier as she works to redeem her benefits. This policing layers on to the ways Victoria describes already feeling scrutinized by fellow shoppers and grocery store staff. Her visibility as a person of color on welfare extends from a history of distrust that women, especially Black women, using welfare are "taking advantage" of the system and thereby not adopting market logics of efficiency and personal responsibility in their own lives (Nadasen, 2007). This distrust stemmed in part from the popularization of the caricature of the "welfare queen;" a Black woman living beyond her means because of the financial assistance gained from enrolling in welfare programs (Nadasen, 2007, p. 58). Although there is

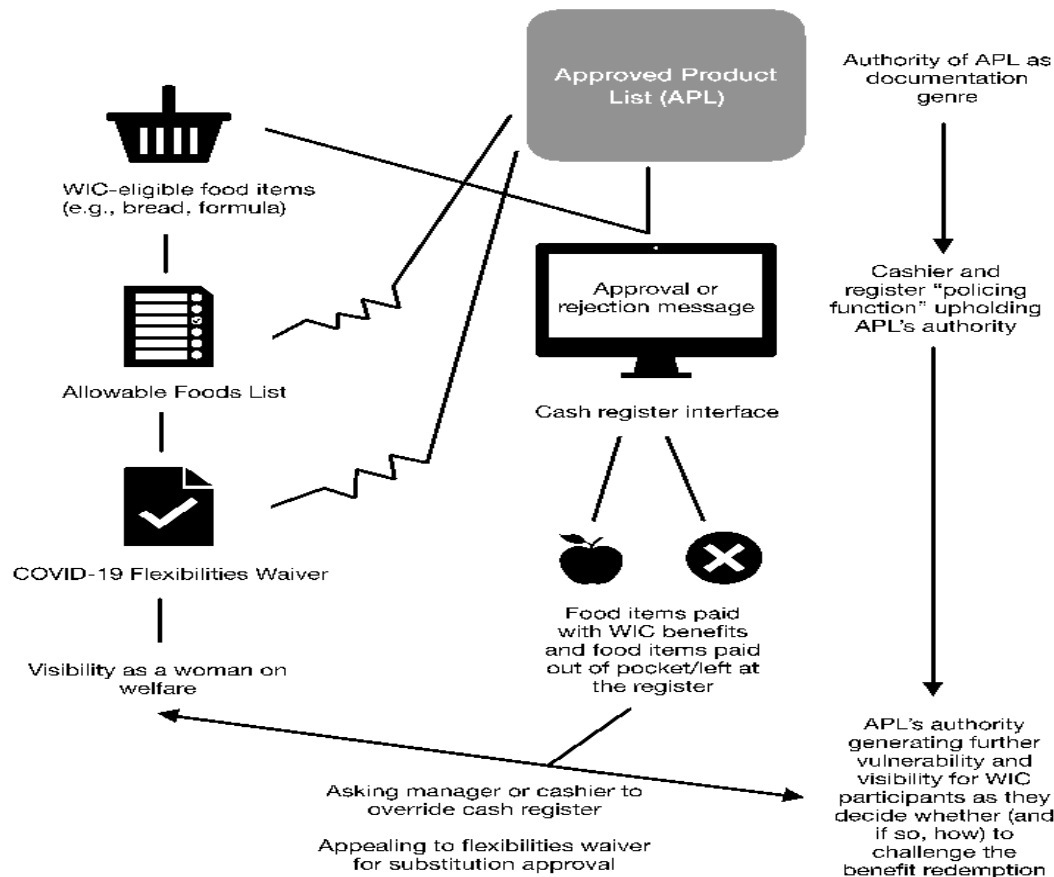


Figure 3: Approved Product List (APL) workflow

little statistical evidence that women and caretakers enrolled in WIC are engaged in fraudulent uses of the program, an underlying fear of and reaction to fraud is built into WIC's information infrastructure, in part through the way the APL hyper-standardizes eligibility and creates such a high-stakes checkout experience. And though all women and caretakers enrolled in WIC experience this information infrastructure, the stereotype of the "welfare queen" and a basic question of whether women of color are worthy or deserve welfare are significant historical factors that contribute to this infrastructural design. The APL, as part of WIC's information infrastructure, has these assumptions, ideologies, and stereotypes baked into its design. Although the APL is never visible to Victoria, her experience of WIC's infrastructure is a felt experience of these inequities.

The APL's role in WIC's information infrastructure contributes to Victoria's stress about her heightened visibility and vulnerability during checkout. As she explains, "I think, so what happens now? It's like an unknown force that controls my experience. What do I have to do? It feels very different from store to store, even trip to trip. Every month has a new surprise waiting for me in the grocery store ... we have times where we can use WIC and times where we go without."

Lavender

Like Victoria, Lavender described feeling hyper-visible during shopping and checkout. As a nonbinary, disabled, pregnant person, Lavender ended up choosing to avoid going into the grocery store altogether and opted for curbside pickup, which is overwhelmingly unavailable to most WIC participants. Lavender lives in rural New Hampshire and has only two WIC vendor grocery stores within an hour's drive.

Lavender describes that they "used to go to the store, but I got so angry and anxious just walking in" because the process was so overwhelming and required fast-paced decision-making. Lavender explained that they have "executive function trouble," which makes grocery shopping take longer, and especially so with WIC. Combined with their heightened visibility as a self-described "not super feminine" pregnant person, the fatigue of pregnancy, and the stress and additional health risks introduced by the pandemic, grocery shopping in person wasn't only inconvenient, it was nearly impossible. Lavender explains that they want to remain as "inconspicuous as possible" as someone on WIC, but also someone who is visibly disabled and pregnant, and described this as the primary motivation for getting pickup. In our interview, Lavender explained that instead of shopping in-person at their nearby store, they drive thirty minutes away for a local chain that has implemented curbside pickup for WIC and Supplemental Nutrition Assistance Program (SNAP) participants:

They're one of the only grocery stores that offer it. You shop online through their app and when you checkout you select that you're going to use EBT. Then, you don't have to prepay for it. You submit your order and get notice your groceries are ready 3-4 days later at a specific time. When you get there, the attendant takes your WIC card and debit card and then they run it that way. They ask for your PIN and they go inside and run your card. They offered that I can go inside and enter my own PIN but I'm fine with it. Then they run my debit card as a credit card.

I have to drive thirty minutes from my house to do this, so I could go to a store ten minutes away, but they don't do curbside. And since I'm immunocompromised, I'm not going into stores, my partner isn't going into stores. We plan our grocery trips way in advance, almost a week, so I just use the store app to order what I want. If it's in stock, great, if it's not, oh well.

Through this curbside pickup process, Lavender avoids being physically present during checkout altogether, but enables the APL to exert authority to an even greater extent, because they are not present to negotiate or challenge the transaction as it is processed by a cashier. Specifically, Lavender describes that they "find out things are ineligible after [they] pay ... which is kind of shitty." For example, if Lavender orders pepper jack cheese and selects "WIC payment" in the app, goes on the drive to get the groceries 3-4 days later, and gives the attendant their WIC card and backup debit card, they will find out pepper jack cheese is not WIC eligible (not listed on the APL) and that they've paid out of pocket for the item after the transaction has already been processed, when the attendant hands them their receipt.

Lavender works to establish invisibility in order to mitigate some of the fatigue and heightened visibility that would likely result from cash register interactions. The processes required for successful benefit redemption by shopping in person do not align with Lavender's body, nor their desire to remain as "inconspicuous" as possible as a WIC shopper, but grocery pickup also introduces new problems, as well as limited benefit redemption. In a recent article, Konrad (2021) theorized access fatigue, the labor and the exhaustion disabled people experience as they create moments of access for themselves. Konrad noted that the "sheer volume and frequency of the rhetorical work of access" cannot "be isolated from other rhetorical burdens that result from oppression and marginalization" (p. 193). This fatigue resonates specifically with Lavender's experience, a nonbinary, disabled, and pregnant person who expends time and energy to meal plan weeks ahead, drive to an inconvenient grocery store because it allows curbside pickup, and accept quietly whatever items are accepted or rejected as WIC-eligible in order to both advocate for themselves (to get the food they need) as politely and invisibly as possible. Therefore, the APL's authority is fully exerted on Lavender because they have opted for a benefit redemption process that prioritizes their well-being over one that prioritizes spending as little money out-of-pocket as possible.

Lilah

Lilah, a white woman living in New Jersey with her husband, Jackson, and two children, did not report experiencing any hyper-visibility, discomfort, or feelings of being scrutinized by cashiers or managers during her descriptions of her experiences redeeming her

WIC benefits. Instead, Lilah reported that meal planning, selecting items at the grocery store, and understanding her WIC benefits to be "relatively smooth-sailing" and that people paid little notice. However, like Victoria, Lilah experiences consistent issues at the cash register when attempting to check out. Between unhelpful cashiers who haven't completed WIC transaction training, to frustrated customers behind her, using WIC was an "upsetting, embarrassing time, every time." Most often, at least one item she knew was WIC approved was denied.

Even though she had purchased the item before using her benefits, another store would deny the item (an indicator that the item is not documented on that store's most recently downloaded APL). Lilah described the full process of WIC benefit redemption as "humiliating, time-consuming, and stressful." She said that she often asked herself "Is it worth it? Can I make it through without using this check, this day?"

However, Lilah developed an innovative strategy to mitigate these frustrating checkout experiences after having an unusual experience with a cashier:

I grabbed the wrong milk. I went up to the register, waited in line, and then realized. I apologized and asked the cashier if she could run someone back to go get the correct milk. And she's like, 'Oh, I'll just substitute. I'll override.' And I was like, 'What?' And she's like, 'It's not a big deal.' But I told her up until the point it really has been a big deal, up to this point nobody has substituted anything for me. And I left with almost no groceries. And she said, 'Oh, well whoever told you that is wrong. We've been told to substitute to get you want you need.' So I wrote her name down, and decided I was going to get her work schedule because all the other cashiers and supervisors that I asked said no.

From this point forward, Lilah called ahead to get "Callie's" work schedule, and only grocery shopped when she was working. Callie was resourceful, solutions-oriented, and as Lilah describes, developed a "rapport" with her that felt supportive. Grocery shopping around Callie's work schedule was not convenient, and was only possible when it aligned with Lilah's work schedule, but avoided inexperienced cashiers and frustrated fellow shoppers.

However, as the pandemic continued, Jackson lost his job, Lilah's hours increased, and Jackson became the point person for grocery shopping and WIC benefit redemption. Callie stopped working at the Wal Mart, and they could no longer rely on her support. During this shift, Lilah describes that it became clear that Jackson experienced far less pushback from cashiers and stigma from fellow shoppers than when she grocery shopped. Unlike Lilah, Jackson challenges cashiers frequently. As he describes,

It's horrendous. I know I'm going to get in line, and I try to get my WIC stuff out all organized, and if you get a young kid [as a cashier], they complain. They say, "Oh, I hate this WIC stuff" ... I'm organized, I'm ready to go. I know my WIC stuff, I got it lined up. Now I'm at the mercy of the register, the cashier. They're like, well, it's not my problem we don't have your WIC stuff. I say, well, it is, because you need to replace it. You have an agreement with the state to provide this, so my understanding is you are required at this point to offer a substitute.

Often when Jackson challenges the cashier, they yield and make the substitution, or find a cashier or manager who knows how to process the transaction. Thus, Jackson creates his own loophole of sorts and circumvents the authority of the APL as a genre by leveraging his authority and privilege. Jackson feels more comfortable with confronting the cashiers, even though it's an unpleasant experience. As a white man redeeming WIC benefits, he experiences some of the same challenges as Lilah, but seems to be able to more readily redeem their WIC benefits because he is not fazed by the inviting conflict and challenging the cashier and extending the time of the transaction. His privilege seems to protect him from the hyper-visibility that WIC participants like Victoria and Lavender experience, as well as the humiliation that Lilah describes when it becomes apparent she is using WIC. Thus, Jackson's experience of WIC's information infrastructure, and the influence of the APL, is markedly different from that of the other participants interviewed for this study. Jackson appears to leverage his privilege with cashiers, demanding that the WIC vendor grocery stores uphold their end of the bargain, rather than accepting the outcome of a WIC transaction to remain inconspicuous or avoid scrutiny. Here we can see how the assumptions, ideologies, and stereotypes baked into the APL as part of WIC's information infrastructure do not particularly marginalize or harm Jackson. Ultimately, Lilah benefits from Jackson's assertion of his privilege because of a net increased benefit redemption, but this inequitable treatment is damaging for Lilah; she is concerned about what will happen when Jackson begins to work again full time and she has to redeem WIC benefits again without the help of Callie.

Harmony

As her self-chosen pseudonym suggests, Harmony's experience with WIC differs distinctly from every WIC participant interviewed so far from this study: it is relatively harmonious. Harmony is a Black woman living in rural Kansas, with access to only one WIC vendor grocery store, but to date, has not experienced any major problems when redeeming her WIC benefits. Although Harmony describes grocery shopping during the pandemic and using WIC as "stressful" because of the lack of pickup or delivery options, overall, Harmony says she "can get whatever she needs" when she shops. Checkout takes longer, but she "hasn't had any issues with things not being in stock or things not scanning correctly." Cashiers seem to be consistently trained and know how to process WIC transactions, are polite, and fellow shoppers don't complain or ask her invasive questions, like Victoria. Harmony did not report feeling hyper-visible as a Black woman on welfare, nor did she report that she feels embarrassed or stigmatized by others during shopping or checkout.

Harmony's experiences with WIC highlight how experiences of information infrastructure operationalize uniquely at an individual level—is her benefit redemption unproblematic because of a consistently accurate APL? Yes, and it also seems that knowledgeable and polite cashiers and managers, fellow shoppers who don't create a scene when in line, and a generally well-stocked and organized grocery store all layers on to the APL's functionality to support and generate a positive overall experience. If the APL began to be inconsistently pulled down to registers (perhaps because of a new, inexperienced manager), or if WIC items went unstocked, or if a new wave of cashier hires created a negative culture of stigmatizing WIC participants, Harmony's infrastructural experiences could change dramatically.

Through mapping the APL utilizing infrastructural inversion, this study found that the APL creates a hyper-standardized and high-stakes environment during checkout at the grocery store cash register, creating a shared set of problems for program participants:

- 1) The APL, although not overtly present during checkout, pits program participants and cashiers against one another in the event of a failed transaction. The failure or success message on the cash register, relayed by the cashier, is authoritative, and, program participants must choose strategically when to challenge this authority.
- 2) The over-reliance of APL documentation to certify or deny food item eligibility can undermine a program participants' knowledge and expertise of what they can purchase with their WIC benefits, resulting in discouragement and limited benefit redemption.
- 3) The high-stakes environment at the cash register creates not only inconvenience for program participants, but can create a vulnerable/unsafe environment for marginalized individuals, including BIPOC, disabled people, and LGBTQIA+ people.

Although preliminary, these findings are significant because they demonstrate the influence of information infrastructure genres in shaping access for program participants in WIC. Although the APL is certainly pragmatic (a necessary document for EBT functionality, an important documentation site), it affects the day-to-day lives of WIC program participants, and not equitably.

IMPLICATIONS

Understanding how the APL mediates and can generate high-stakes negotiations during checkout clarifies how genres that make up information infrastructures can significantly shape access to resources, like the food WIC participants should be able to purchase using their government benefits. Even without direct knowledge of the presence of the APL, WIC participants must choose how and when to advocate for themselves in the hyper-visible environment of checkout, where cashiers and other customers draw their attention to the nature of a participant's payment method. These WIC participants carry all responsibility to choose the correct items, yet they have little agency to challenge item eligibility at the point of sale, highlighting that the EBT system contributes to a WIC grocery shopping experience that relies on the APL as a primary authority, regardless of how accurate it is. This seems to hold WIC participants to an impossible standard—to be personally responsible in their food item selection, following the rules exactly in order to optimize their benefit redemption, and to submit to the authority of the APL without question.

This study's findings have implications for studying information infrastructure genres and the ways writing studies research might take up greater attention to infrastructures in qualitative approaches to locate sites of inequity. The APL is a genre which produces a relationship between cashier, cash register, and WIC participant that heightens visibility to bodies that may already be more visible (Black, disabled, pregnant) and, therefore, already subject to unwanted attention, surveillance, and/or policing. Ultimately, this study found that APLs do not affect WIC participants equitably. What other genres that function as part of information infrastructures might also inequitably affect access? What "systems" contain these oft-unnoticed, invisible genres?

It is hard work to be on government assistance—especially in the pandemic, but to an extent always—even if it works. If it doesn’t work, that’s even more work, and work that tends to place blame on participants, increase their visibility in a process that can already be stigmatizing, and wear on their bodies. Studies of WIC participant shopping experiences (Bertmann et al., 2014; Gleason et al., 2014; Najjar, 2013; Todd et al., 2010) have recognized that negative grocery store interactions result not only in embarrassment and stigmatization, but lowered program participation rates and high attrition. For example, Bertmann et al. found that “barriers,” including “experiences of negative interactions in stores, dealing with misunderstandings over WIC benefit redemption rules, or feeling embarrassed or judged in relation to using WIC benefits” (p. 56) are directly related to an under-redemption of benefits. Bertmann et al. suggested that understanding how “more experienced WIC participants make use of . . . benefits and whether these strategies could be taught to new WIC participants as they enter the program” (p. 57) could be useful. In part, this study suggests that there is much to be learned from the workarounds, loopholes, and technical labor that people perform to create access for themselves as they navigate the genres of information infrastructures.

Without the context of the APL’s role as part of WIC information infrastructure, a genre analysis of WIC benefit redemption genres and the recurrent situation of stymied benefit redemption may uphold the narrative that WIC participants are ultimately at fault when their transactions fail. A genre analysis without this infrastructural context might position the error message present on the cash register interface as authoritative, uncomplicated—it says the item isn’t WIC eligible, so it must not be. WIC participants must be selecting incorrect items from the shelves and trying to pay for them with their benefits. This genre analysis might conclude that WIC participants engage in strategies to create access for themselves because the rules of the program are confusing—which is why they’re selecting the wrong items, for instance. An analysis of WIC participant’s benefit redemption texts and interviews with WIC participants would uphold the APL’s role as an invisible, and ultimately unnoticed, genre.

Adding an infrastructural perspective to genre analysis adds another lens to studying the world through “genre-colored glasses” (Devitt, 2015) that enriches our study of genre access. These genre-infrastructure glasses (I picture them as steampunk glasses) help us to see that the interaction between a WIC participant and a cashier at the cash register is mediated not only by visible resource genres (assigned food package, Allowable Foods List, flexibilities flyer, eWIC card) and EBT technology (cash register, scanner, PIN pad), but also invisible genres that make up the information infrastructure, like the APL. In fact, both the visible resource genres and EBT technology are mediated by WIC’s information infrastructure via the APL (see Figure 1). An infrastructural perspective locates and provides an understanding, then, of why WIC participants need to engage strategies to ensure successful benefit redemption, as well as what these practices actually address in terms of the disconnect between the local and global in supporting social action.

Although we learn (and learn through) genres in order to get things done in the world, genres that make up information infrastructures like the APL seems to complicate things. Miller (1984) described that learning a genre means better understanding a given situation and taking on a particular orientation towards what is possible by developing a “cultural rationality” (p. 165). This study of the APL,

I think, showcases WIC participants’ genre knowledge (drawing on their resource genres) and a kind of infrastructural knowledge. Interestingly, though, this study doesn’t show that WIC participants “learn to understand better the situations in which [they] find [themselves]” (Miller, 1984, p. 165) – they develop sophisticated strategies for navigating “the system” without understanding why what is happening is happening. WIC participants learn how to effectively navigate grocery shopping as they participate in the program. The WIC participants interviewed for this study did not understand why benefit redemption kept going poorly, only that if they wanted to acquire food through WIC, they needed to employ these strategies. Although they drew on the available benefit redemption genres and employed these sophisticated strategies for ensuring successful transactions, this genre action was a form of cultural rationality that responded to a problem, the source of which was unknown.

Because the APL operates truly behind-the-scenes, I’m not sure if it’s accurate to describe participants’ strategies for creating access as a traditional kind of genre knowledge. Instead, this infrastructural (genre?) knowledge seems to be similar, in that it’s deployed strategically and is context-bound (in terms of time and place). In the case of the APL, WIC participants still interact with the text, but the APL mediates the direct interaction with the cashier and cash register. The relationship between the information infrastructure genre (APL) and more “present” genres (participant-cashier negotiations, resource genres like the Allowable Foods List) seems to be mediation, or perhaps more accurately, adjudication. A WIC participant brings food items to the conveyer belt, with evidence of their eligibility (often through the Allowable Food Items List and/or flexibilities flyer), and the APL file adjudicates, makes a final judgement, on whether items are eligible or not. The cashier or manager can act as intercessor (like Callie) who intervenes and “overturns” the decision based on evidence, or can uphold the “law” set out by the APL via the cash register. In this metaphor, WIC participants are, in a sense, “on trial” at the cash register and bear the burden of proof to show food item eligibility, and therefore at times must go to great lengths to guarantee as uncomplicated of a transaction as possible, even though the “law” (the specific APL file downloaded) often differs from store to store. That genres like the APL within information infrastructures may vary widely across large organizations like WIC, therefore shaping individual experience of those organizations significantly, suggests that adding an infrastructural perspective to more traditional genre analysis is needed to get the full picture.

The APL is a “system thing”—as an information infrastructure genre, it both enables and restricts social action; it enables WIC’s EBT system “to work” and at the same time, actively works against many WIC participants, especially so against people already overburdened, under-resourced, and vulnerable. There’s no quick genre “fix” for the problems the APL generates, and no one-size-fits-all intervention into WIC’s information infrastructure that resolves the barriers currently present for program participants. However, better understanding the root of the problem (genres like the APL) within information infrastructures is a good start for writing researchers interested in bureaucratic and institutional critique and intervention.

LIMITATIONS

The COVID-19 pandemic was both a limitation and opportunity to study how a genre, the APL, as part of a wider information
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infrastructure, shapes people's access to fully redeeming their government benefits and purchasing food, and upholds and perpetuates inequity. Though I was able to hear from a wide range of WIC participants and hear their stories, ideally, the study would have benefitted from recruitment methods beyond convenience sampling. Additionally, it would have been useful to grocery shop alongside program participants to observe interactions with cashiers and participants in real time. Overall, however, convenience sampling did provide a way to understand the wide spectrum of experiences of program participants across different states. Ultimately, more research is needed to further understand the role of the genres like the APL that make up information infrastructures, not only within bureaucratic contexts like government benefit programs, but across academic, public, and workplace contexts.

CONCLUSION

The genres that make up information infrastructures work invisibly to shape people's access to resources. In the case of WIC, the hyper-standardized environment at checkout generated by the APL's function as an information infrastructure genre created problems for WIC program participants who were already more visible and vulnerable. As a genre that is "behind-the-scenes" in WIC's information infrastructure, the APL has serious consequences for benefit redemption, and does shape the experience of WIC program participants as they attempt to get the groceries they need. The APL is one example of a genre within one large organization's information infrastructure that affects people's well-being, dignity, and health, and writing studies and technical communication as a field should turn greater attention to these "behind-the-scenes" genres within information infrastructures. Federal programs like WIC, the Supplemental Nutrition Assistance Program (SNAP), and Temporary Assistance for Needy Families (TANF) all may be beneficial sites to study to further understand how the genres that make up information infrastructures influence despite their invisibility. Pursuing this kind of infrastructural analysis would provide more insight into the relationships between writing, infrastructures, and access and further develop the growing "infrastructural turn" within the field of writing studies.

As a result of these findings, scholars in writing studies and technical communication studying writing infrastructures might also consider the following:

- When collecting genre samples for analysis from an organization, consider whether there are relevant samples that derive from a site's information infrastructure (government programs like WIC have these publicly available on their website)
- If conducting in-person observation of genre interactions, review the infrastructural functions of genres that are unavailable to observation, but may influence these interactions
- Pause before discarding technical documents that resist direct analysis

Ignoring the genres that make up information infrastructures risks identifying the symptoms of a problem without locating an underlying cause. Locating and analyzing the genres that make up information infrastructures is one way to "drill deep into ... the infrastructure of inequality" (Grabill, 2010, p. 455) of complex systems and large organizations.

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